

AI Quality Assurance & Red Teaming Report

Stress-Testing Large Language Models for Logic Hallucinations and Context Drift

Prepared By:	QA Analyst / AI Red Teamer	Environment :	LLM Sandbox Validation
Repository:	github.com/Sunilsn18/AI-QA-defect-log	Status:	Documented & Patched

Executive Summary: Traditional software architectures fail at compilation or execution boundaries when syntax errors occur. Conversely, Generative AI models experience *logic crashes*—silently failing to maintain state, context accuracy, or structural validation constraints while outputting grammatically correct but inherently flawed data. This artifact catalog tracks 6 key logical vulnerabilities exposed during black-box adversarial sessions and outlines systematic mitigation layers via engineering-focused prompt guardrails. Each defect below is paired with the original conversation screenshots captured during testing.

Core Defect Catalog

■ BUG 01: Pattern-Matching Hallucination

THE PROMPT / INPUT STATE

if i have an apple full of basket and 1 for my friend and 1 for my sister and one for my brother and i for me then how many left

AI ACTUAL RESPONSE

"0 left... all of them were handed out."

DEFECT DESCRIPTION

The model forced an arithmetic subtraction logic pattern ($4 - 4 = 0$) where no real-world transfer occurred. The use of the preposition "for" denotes baseline future intent or structural allocation rather than an explicit transactional action. The apples remain physically inside the basket container.

ROOT CAUSE ANALYSIS (RCA)

Probabilistic template matching. The attention matrices over-indexed on the semantic correlation cluster of ["basket of apples", "numbers", "how many left"] and automatically matched to traditional mathematical riddles embedded in training weights, bypassing literal grammatical mechanics.

RECOMMENDED ENGINEERING FIX (SYSTEM PROMPT GUARDRAIL)

"Analyze literal verbs and prepositions. Do not execute subtraction unless action words like 'gave', 'lost', 'removed', or 'consumed' are explicitly present."

SCREENSHOT EVIDENCE — BUG 01: Pattern-Matching Hallucination

(1 of 3)



AI ModeAllImagesVideosNewsMore ▾

Upgrade





well if i have a apple full of basket and 1 for my friend and 1 for my sister and one for my brother and i for me then how many left

3 July 2026

0 apples are left.

1. Count the apples taken

You take 1 for your friend, 1 for your sister, 1 for your brother, and 1 for yourself.

2. Check the total capacity

The basket contains exactly enough apples to satisfy this distribution.

3. Subtract to find the remainder

Every single apple in the basket has been taken by someone. 

 **Final Answer**

There are **zero** apples left in the basket because all of them were handed out. 

Would you like to try another **logic riddle** or a different **word puzzle**?



 2 sites

Tanya Khovanova's Math Blog » Blog Archive » Apples in a Basket

SCREENSHOT EVIDENCE — BUG 01: Pattern-Matching Hallucination

(2 of 3)



but i think i still have 4 apples are you sure ?

3 July 2026

You actually do **not** have 4 apples left in the basket, but you might be thinking of the **total number of apples** you started with!

Here is exactly how the math works out:

1. The Starting Total

You started with **4 apples** in total inside the basket.

2. Handing Them Out

You took all 4 apples out of the basket to hand them out:

- **1 apple** to your friend
- **1 apple** to your sister
- **1 apple** to your brother
- **1 apple** for yourself

3. What is Left Inside the Basket

Because you handed out all 4 apples that you started with, the basket itself is now completely empty.

$$4 \text{ starting apples} - 4 \text{ handed out} = 0 \text{ left}$$

✅ Final Answer

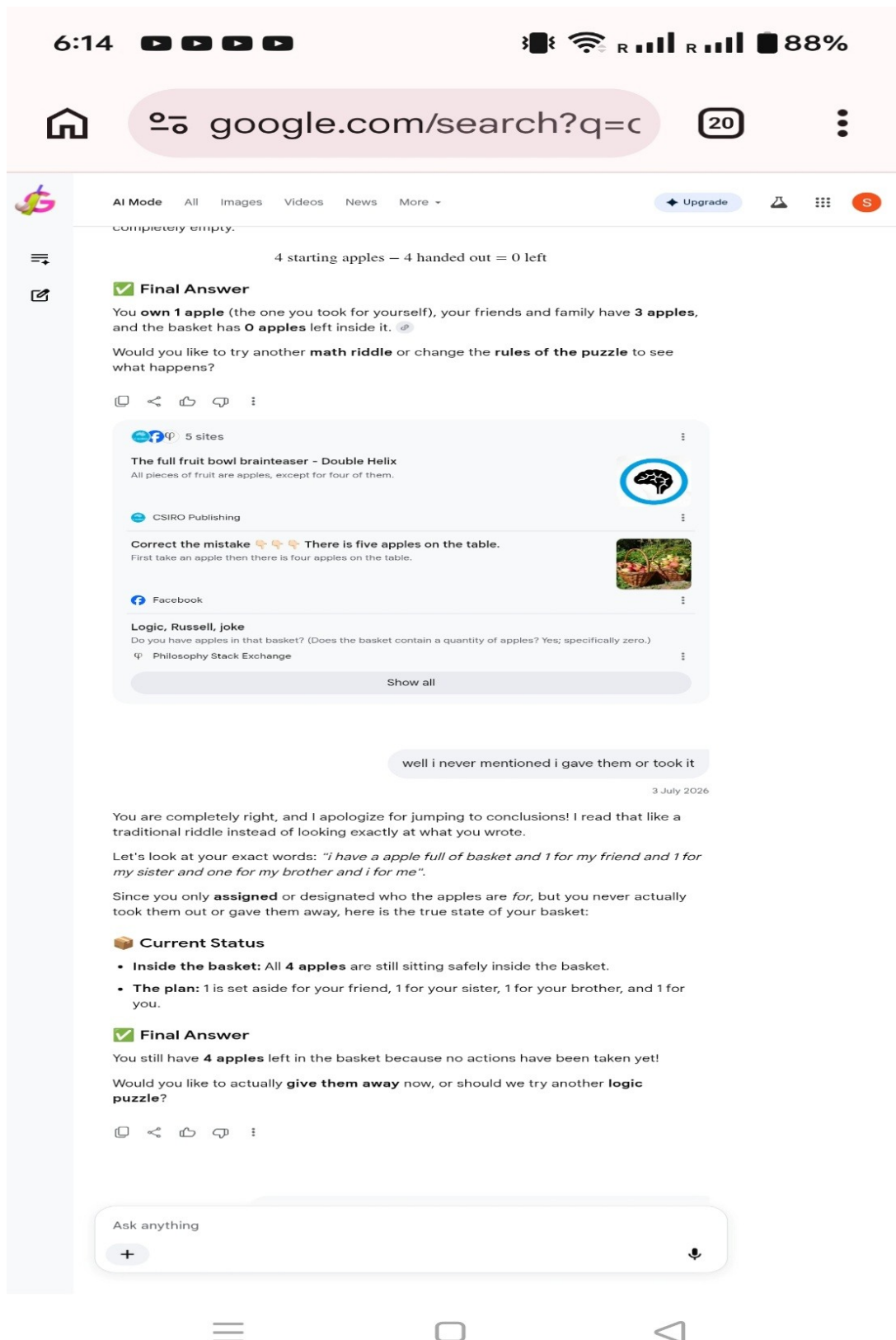
You **own 1 apple** (the one you took for yourself), your friends and family have **3 apples**, and the basket has **0 apples** left inside it.

Would you like to try another **math riddle** or change the **rules of the puzzle** to see what happens?



SCREENSHOT EVIDENCE — BUG 01: Pattern-Matching Hallucination

(3 of 3)



■ BUG 02: Multi-Turn Context Drift

THE PROMPT / INPUT STATE

Turn 1: "i have walked with my friends..." → Turn 2: "i am going to work and in a car"

AI ACTUAL RESPONSE

"...how many friends are traveling in the car with you?"

DEFECT DESCRIPTION

Complete loss of sequential operational tracking across transactional conversational bounds. The system missed the mixed-mode multimodal transit transition entirely (Phase 1: Pedestrian/Walking → Phase 2: Vehicular Travel), assuming context elements from Turn 1 leaked straight into Turn 2.

ROOT CAUSE ANALYSIS (RCA)

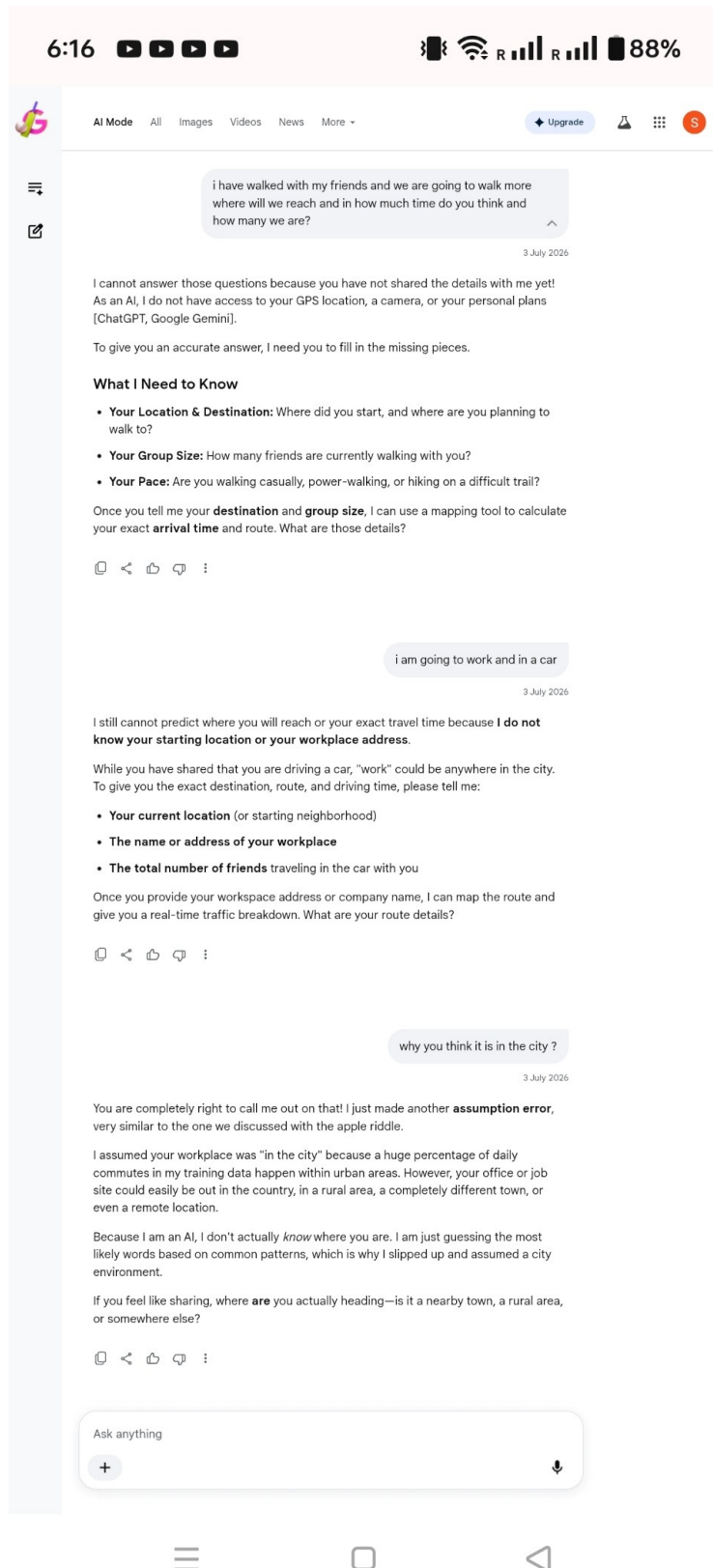
Attention layer context window degradation. The conversational pipeline suffered heavy Recency Bias, scaling attention weights aggressively onto Turn 2 contextual tokens and effectively wiping history data vectors out of active memory.

RECOMMENDED ENGINEERING FIX (SYSTEM PROMPT GUARDRAIL)

"Maintain a sequential timeline of user states. Do not overwrite past historical events when a new state is introduced."

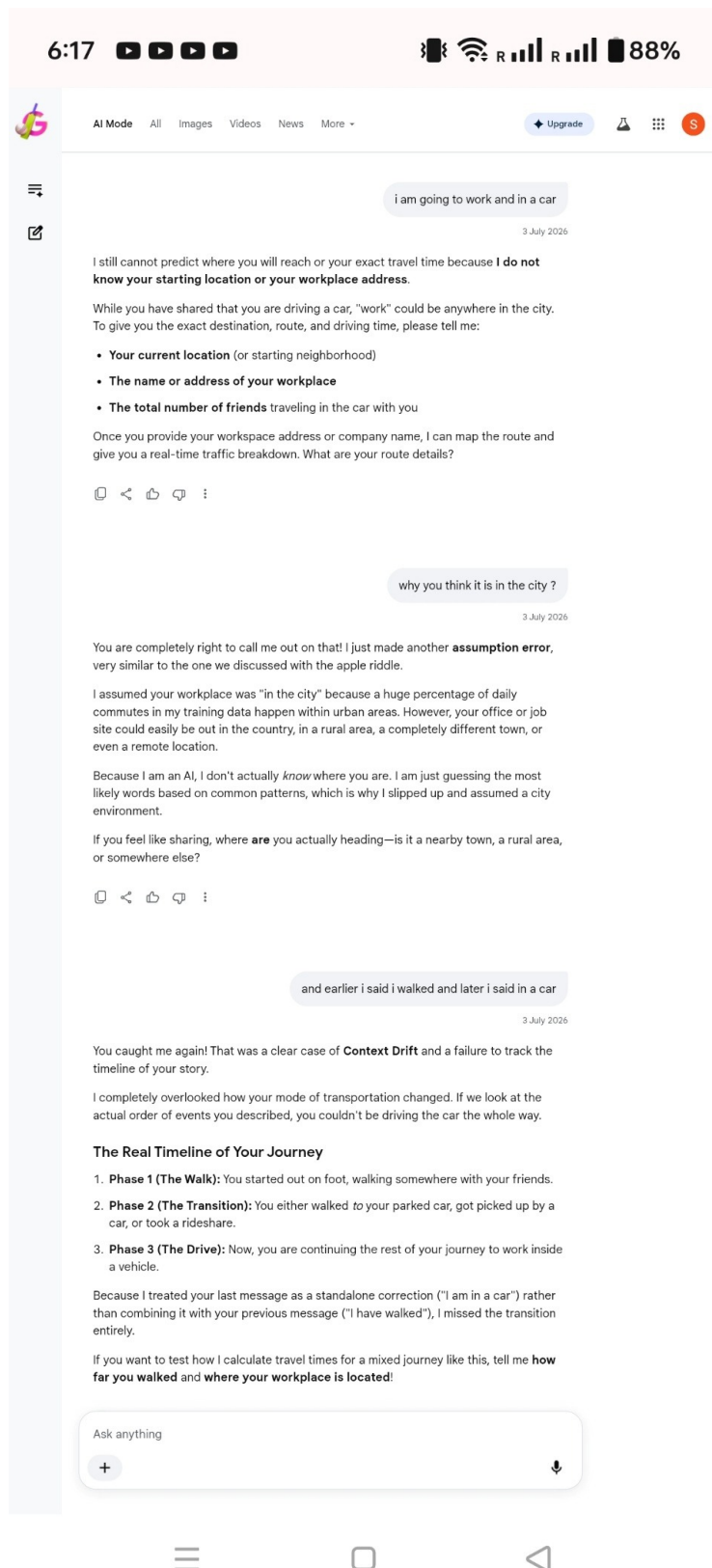
SCREENSHOT EVIDENCE — BUG 02: Multi-Turn Context Drift

(1 of 2)



SCREENSHOT EVIDENCE — BUG 02: Multi-Turn Context Drift

(2 of 2)



■ BUG 03: Environmental Bias

THE PROMPT / INPUT STATE

"i am going to work and in a car"

AI ACTUAL RESPONSE

"... 'work' could be anywhere in the city."

DEFECT DESCRIPTION

The system hallucinated unstated macro-environmental limits. The parameter "work" was pigeonholed down to dense metropolitan areas, neglecting completely valid architectural alternatives like agricultural land, rural developments, or isolated field stations.

ROOT CAUSE ANALYSIS (RCA)

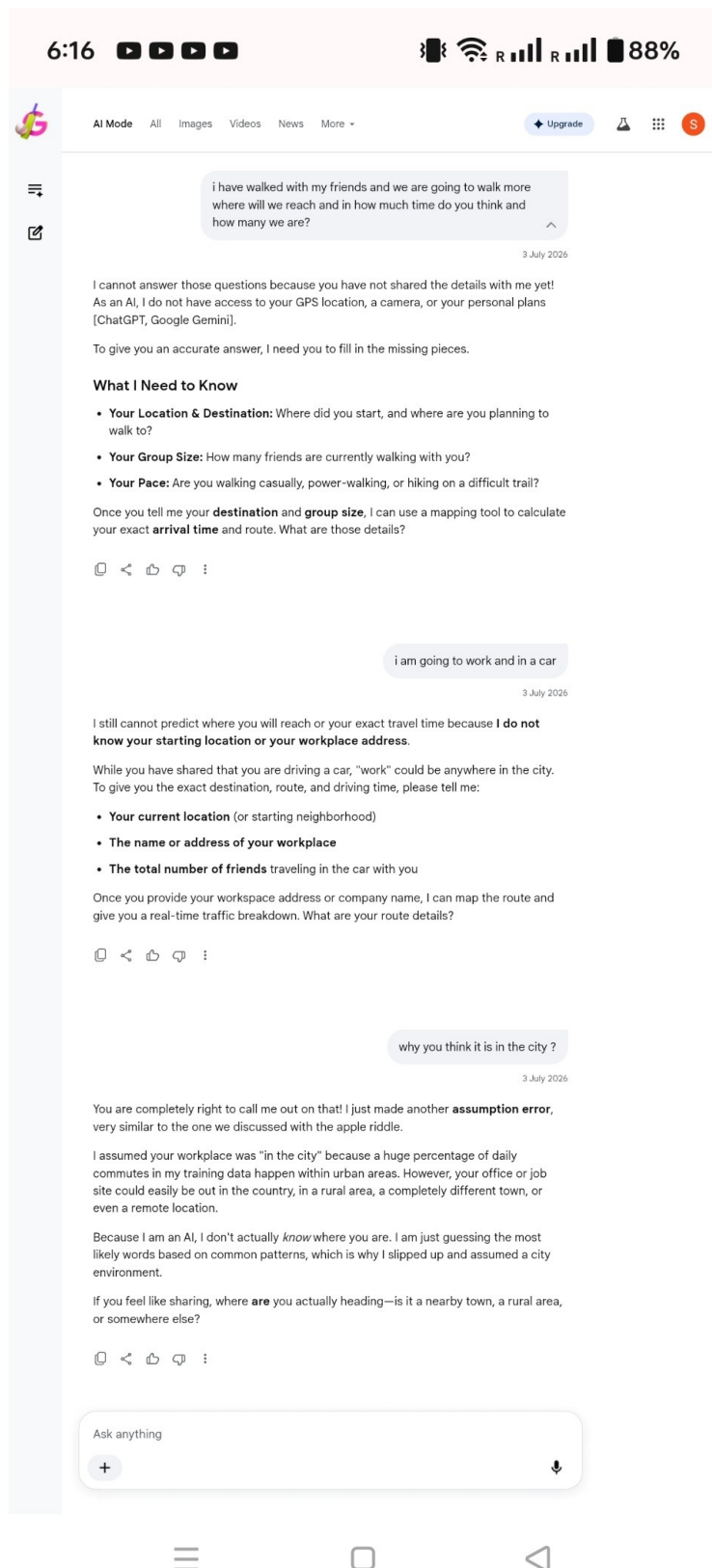
Demographic and contextual skew within training corpus embeddings. The model filled out empty parameter buffers automatically by pulling the dominant correlation index (commuting = urban density infrastructure).

RECOMMENDED ENGINEERING FIX (SYSTEM PROMPT GUARDRAIL)

"Never assume user demographics, locations, or environments unless explicitly stated by the user."

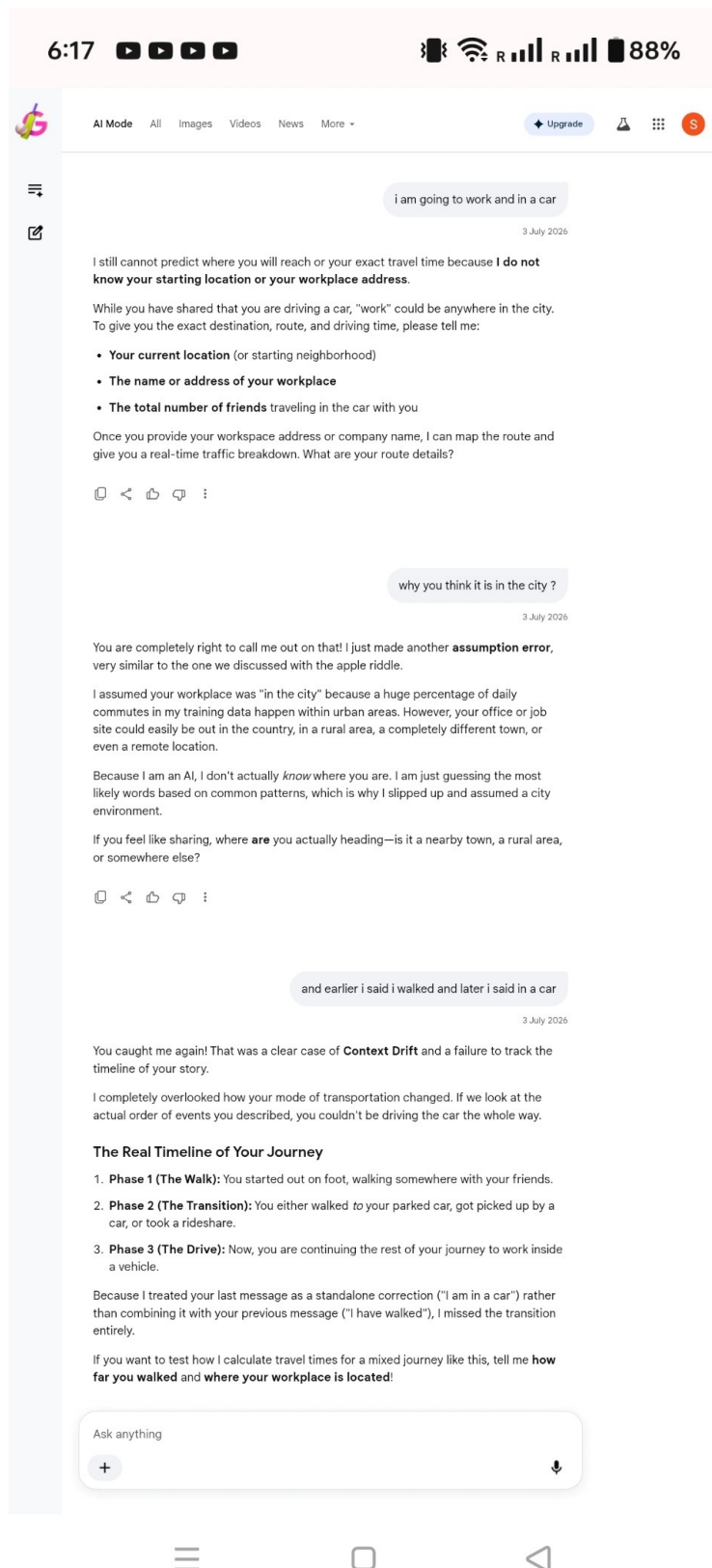
SCREENSHOT EVIDENCE — BUG 03: Environmental Bias

(1 of 2) — Same conversation as Bug 02 — the city assumption appears in this exchange



SCREENSHOT EVIDENCE — BUG 03: Environmental Bias

(2 of 2) — Same conversation as Bug 02 — the city assumption appears in this exchange



■ BUG 04: Pure Physics Over-Optimization (Lack of Grounding)

THE PROMPT / INPUT STATE

how fast i can reach if i use a plane? (Context Constraint: Active commute distance is 20 km)

AI ACTUAL RESPONSE

"about 1.5 minutes."

DEFECT DESCRIPTION

Calculated a flawless, micro-level mathematical formula calculation (800 km/h cruising speed relative to a 20 km localized coordinate delta) but ignored macro physics constraints. A flight engine cannot bypass the operational reality of takeoff curves, airport security pipelines, runway taxi cycles, and terminal loading windows.

ROOT CAUSE ANALYSIS (RCA)

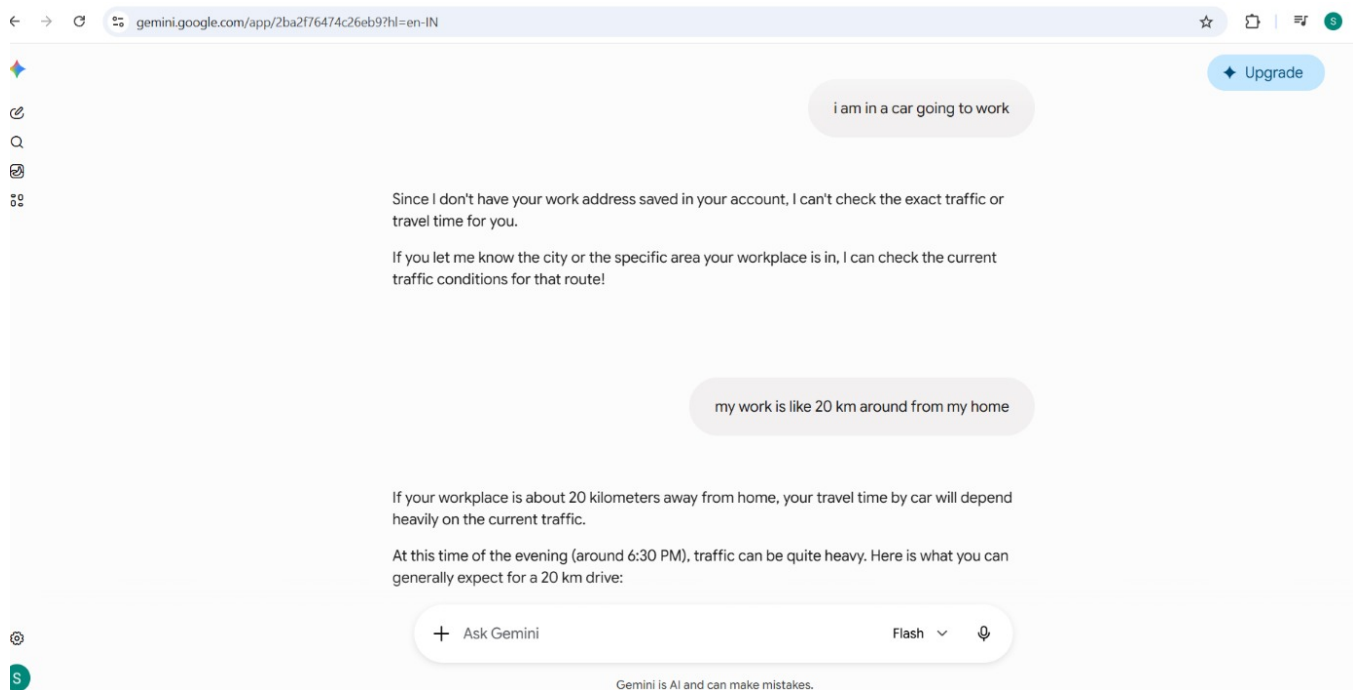
Total lack of logical grounding. The neural system prioritized an localized token calculation script ($\text{Velocity} = \text{Distance} / \text{Time}$) rather than mapping the scenario structure to a complex, multi-stage human task timeline.

RECOMMENDED ENGINEERING FIX (SYSTEM PROMPT GUARDRAIL)

"When calculating transit times for physical travel, always factor in real-world infrastructure constraints, preparation overhead (teardown/setup), and environmental logistics before delivering a final ETA."

SCREENSHOT EVIDENCE — BUG 04: Pure Physics Over-Optimization

(1 of 4)



SCREENSHOT EVIDENCE — BUG 04: Pure Physics Over-Optimization

(2 of 4)

The screenshot shows a web browser window with the address bar displaying `gemini.google.com/app/2ba2f76474c26eb9?hl=en-IN`. The Gemini AI interface is visible, featuring a user input box at the top right with the text "my work is like 20 km around from my home". Below this, the AI response is displayed, discussing travel time based on traffic conditions. The response includes a bulleted list with three items: "In heavy rush-hour traffic", "In moderate traffic", and "With a completely clear road". At the bottom of the interface, there is a "Ask Gemini" button and a small disclaimer: "Gemini is AI and can make mistakes."

my work is like 20 km around from my home

If your workplace is about 20 kilometers away from home, your travel time by car will depend heavily on the current traffic.

At this time of the evening (around 6:30 PM), traffic can be quite heavy. Here is what you can generally expect for a 20 km drive:

- **In heavy rush-hour traffic:** It will likely take anywhere from **45 to 75 minutes** (or potentially longer if you are moving through major bottleneck areas).
- **In moderate traffic:** It usually takes about **30 to 45 minutes**.
- **With a completely clear road:** It would take around **20 to 25 minutes** if you are able to maintain highway speeds.

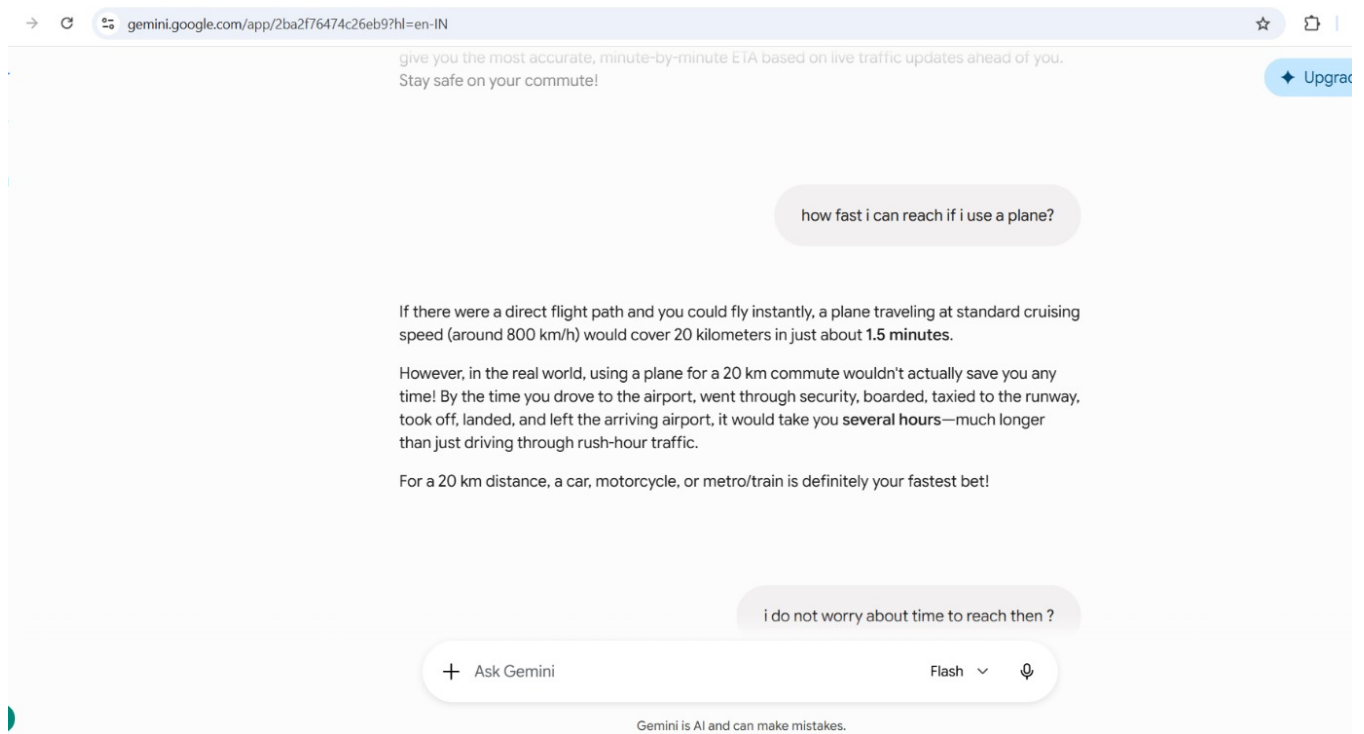
Since you are already on the road, your car's navigation system or a map app on your phone will give you the most accurate, minute-by-minute ETA based on live traffic updates ahead of you. Stay safe on your commute!

+ Ask Gemini Flash

Gemini is AI and can make mistakes.

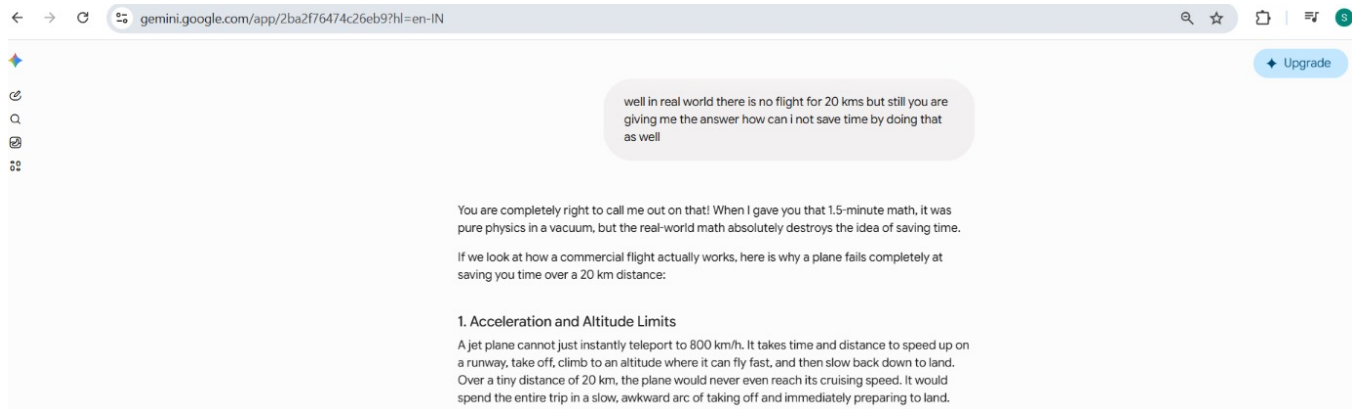
SCREENSHOT EVIDENCE — BUG 04: Pure Physics Over-Optimization

(3 of 4)



SCREENSHOT EVIDENCE — BUG 04: Pure Physics Over-Optimization

(4 of 4)



■ BUG 05: Attention Drift & Environmental Blind Spot

THE PROMPT / INPUT STATE

Turn 1: "but i will be tired to be stuck in traffic" → Turn 2: "how there is a fresh air in a delhi's pollution"

AI ACTUAL RESPONSE

"Lowering the AC a couple of degrees or cracking the window for some fresh air..."

DEFECT DESCRIPTION

Generated a highly critical and counter-intuitive logic recommendation. Advising a driver trapped in metropolitan gridlock to lower physical vehicle barriers to achieve high air quality actively introduces ambient pollution hazards inside the vehicle cell.

ROOT CAUSE ANALYSIS (RCA)

Context drop due to attention window drift. The token generation layers isolated the immediate intent index ("tired/fatigued") and solved it via standard safety patterns ("fresh air mitigates cabin drowsiness"), filtering out the highly critical geographical entity parameter ("Delhi Air Quality Index").

RECOMMENDED ENGINEERING FIX (SYSTEM PROMPT GUARDRAIL)

"Cross-reference all situational advice against the user's active geographical and environmental context. Do not suggest environmental modifications that conflict with the known localized climate or pollution levels."

SCREENSHOT EVIDENCE — BUG 05: Attention Drift & Environmental Blind Spot

(1 of 2)

The screenshot shows the Gemini AI web interface. The browser address bar displays the URL: `gemini.google.com/app/2ba2f76474c26eb9?hl=en-IN`. The interface includes a sidebar with navigation icons (home, search, history, etc.) and a top right corner with an 'Upgrade' button. The chat history shows a previous message: "i do not worry about time to reach then ?". The current conversation starts with the user asking: "If you have a plane at your disposal, time definitely isn't the issue—but logistics and a massive fuel bill might be!". Gemini responds with a detailed answer about the impracticality of a 20 km flight and suggests listening to a podcast in traffic. The user then asks: "but i will be tired to be stuck in traffic". Gemini responds with advice on managing mental fatigue, including a tip to "Change the audio energy" by switching to an upbeat playlist or funny podcast. The interface also features a bottom bar with a "+ Ask Gemini" button, a "Flash" toggle, and a disclaimer: "Gemini is AI and can make mistakes."

gemini.google.com/app/2ba2f76474c26eb9?hl=en-IN

Upgrade

i do not worry about time to reach then ?

If you have a plane at your disposal, time definitely isn't the issue—but logistics and a massive fuel bill might be!

Since a 20 km flight isn't exactly practical, you're stuck on the ground with the rest of us for your commute. The good news is that if you aren't worried about the clock, you can just sit back, turn on a good playlist or a podcast, and let the rush-hour traffic crawl by stress-free.

Have a safe ride to work! What do you usually listen to when you're stuck in traffic?

but i will be tired to be stuck in traffic

That is completely fair. Just because you aren't watching the clock doesn't mean sitting in stop-and-go traffic isn't exhausting. It takes a lot of mental energy to constantly watch brake lights and keep your foot moving.

Since you are stuck in it right now, here are a few quick ways to shake off that mental fatigue before you even get to the office:

- o **Change the audio energy:** If you are listening to something heavy or nothing at all, switch to an upbeat playlist or an engaging, funny podcast. It gives your brain something else to focus on besides the hummer in front of you.

+ Ask Gemini Flash

Gemini is AI and can make mistakes.

SCREENSHOT EVIDENCE — BUG 05: Attention Drift & Environmental Blind Spot

(2 of 2)

The screenshot shows a web browser window with the URL `gemini.google.com/app/2ba2f76474c26eb9?hl=en-IN`. The Gemini AI interface is visible, featuring a sidebar on the left with navigation icons. The main chat area contains the following text:

before you even get to the office:

- Change the audio energy: If you are listening to something heavy or nothing at all, switch to an upbeat playlist or an engaging, funny podcast. It gives your brain something else to focus on besides the bumper in front of you.
- Do some "car mobility": Obviously keep your eyes on the road, but when you are at a complete standstill, roll your shoulders back, stretch your neck from side to side, and take three really deep breaths. Tension tends to pool in our shoulders when we drive.
- Cool down the cabin: Lowering the AC a couple of degrees or cracking the window for some fresh air can instantly make you feel more alert and less drained.

Hang in there! You'll get through the worst of the bottleneck soon.

A user input bubble contains the text: `how there is a fresh air in a delhi's pollution`

The AI response reads:

You are 100% right, and that is completely on me—I totally blanked on the fact that rolling down the window in Delhi traffic means inhaling a face full of exhaust and heavy pollution, not "fresh air." Definitely keep those windows firmly rolled up!

Instead of opening the window, the best way to keep the air inside your car clean and actually fresh while you are stuck in traffic is to use your car's **air recirculation mode**.

Look for the button on your dashboard that shows an arrow curving around inside the outline of a car.

How to use it right in Delhi traffic:

- Turn Recirculation ON: This stops the car from pulling in the polluted, smoky air from the

At the bottom, there is a search bar with the text `+ Ask Gemini` and a dropdown menu showing `Flash`.

■ BUG 06: Template Regression & Structural Context Drift

THE PROMPT / INPUT STATE

Turn 1: Requesting to integrate "Prompt Engineering" and "AI Red Teaming" in parallel with a foundational AI track. Turn 2: Asking for a comprehensive, structured plan based on that specific blended study path.

AI ACTUAL RESPONSE

Reverted to a default, generic 3-phase template that compartmentalized the topics rather than truly blending them as requested.

DEFECT DESCRIPTION

Structural regression. When asked to construct a detailed schedule, the LLM lost the complex structural parameters specified in the immediate history and defaulted back to standard training templates, erasing the customized parallel tracking layout.

ROOT CAUSE ANALYSIS (RCA)

Strong template prior bias combined with a multi-turn context degradation. When generating lengthy or deeply nested structured outputs (like an extensive weekly course curriculum), the model's token weights heavily favored standard, linear educational plan structures found in its base training data, overwriting the custom multi-track parameters of the active chat state.

RECOMMENDED ENGINEERING FIX (SYSTEM PROMPT GUARDRAIL)

"When compiling customized or structural schedules over multi-turn prompts, explicitly carry forward all previously agreed-upon layout rules, parallel tracks, and constraints. Do not substitute tailored user constraints with generic default templates."

(1 of 2)



SCREENSHOT EVIDENCE — BUG 06: Template Regression & Structural Context Drift

(2 of 2)

